

Astrophysical Recipes (Second Edition)

The art of AMUSE

Simon Portegies Zwart, Steve McMillan and Steven Rieder

Epilogue

There is no dark side of the Moon really.

—Gerry O’Driscoll

?
In the last ~400 pages we have presented the Astrophysical Multipurpose Software Environment (AMUSE) and motivated its development and use. We have glossed over many underlying physical processes, and tried to stay away from the many mathematical details. Equally, we have barely discussed the technical challenges and algorithms needed to make all this work. Our objectives have been to present an interesting and novel way in which domain applications can be combined at a meta-level in a homogeneous environment. The methods presented allow developers to preserve their software in a stable and homogeneous habitat, yet also provide tools that allow them to couple these codes. In this coupled environment, it is possible to both test the individual parts and address complex multiscale and multiphysical processes.

The methods discussed and explained in this book are not limited to astronomy. We have worked on expanding the methods to long-term weather prediction and oceanography in a project we call OMUSE (Pelupessy et al., 2016b, a), which stands for (as you might guess) the Oceanographic Multipurpose Software Environment. Further extrapolations on the concept make us wonder if similar environments could be developed for simulations in biology (BMUSE) or chemistry (CMUSE). For the latter, we have worked on the implementation of the chemical evolution codes KROME (Grassi et al., 2014) and UCLChem (Holdship et al., 2017). Both are operational, but so far we have not published any results using them in a peer-reviewed scientific journal.

Furthermore, we have been working on developing a general *MUSE with modules for various individual domains, such as A for Astronomy, B for biology, etc. The basic functionality of AMUSE could provide a template for such an endeavor. The development of AMUSE makes for a very interesting first push in

that direction, although we are still far from a global simulation environment that might be utilized for a wide range of physical applications.

Maybe the biggest challenge to the development of such a meta-framework is not technical but psycho-sociological. In our experience, senior researchers tend to stick to the methods and techniques they developed when they finished their PhDs. Graduate students are directed by their supervisors, which usually confines them to familiar methodologies. Postdoctoral researchers are naturally preoccupied with securing a solid foundation for their future careers. One can then wonder if there is an avenue leading to broad acceptance of the ideas presented here. Maybe there is. We have found on a daily basis that AMUSE has enriched our academic repertoire at all levels, from freshman to postdoc. Most importantly, using AMUSE turns out to be a lot of fun, and working with the package is enormously rewarding.

We hope you have enjoyed this book, using the codes, and hopefully developing more scripts and even publishing the results. We hope you will acknowledge AMUSE (and the constituent code references thrust upon you by our software) in any such work. As a final comment, we emphasize that we see AMUSE as a community effort. In that respect, we hope that any scripts you develop and any software you write will be made freely available to everyone in the field. The AMUSE team will be happy to work with you to incorporate your work as part of the project.

References

- Grassi, T., Bovino, S., Schleicher, D. R., et al. 2014, [MNRAS](#), **439**, 2386
- Holdship, J., Viti, S., Jiménez-Serra, I., Makrymallis, A., Priestley, F., et al. 2017, [AJ](#), **154**, 38
- Pelupessy, I., van Werkhoven, B., van Elteren, A., et al. 2016a, The 5th Int. Workshop on Multi-scale (Un)-structured Mesh Numerical Modeling for Coastal, Shelf, and Global Ocean Dynamic (Delft: TU Delft) 1
- Pelupessy, I., van Werkhoven, B., van Elteren, A., et al. 2016b, 12th International Conf. on e-Science (e-Science) (Piscataway, NJ: IEEE) 399